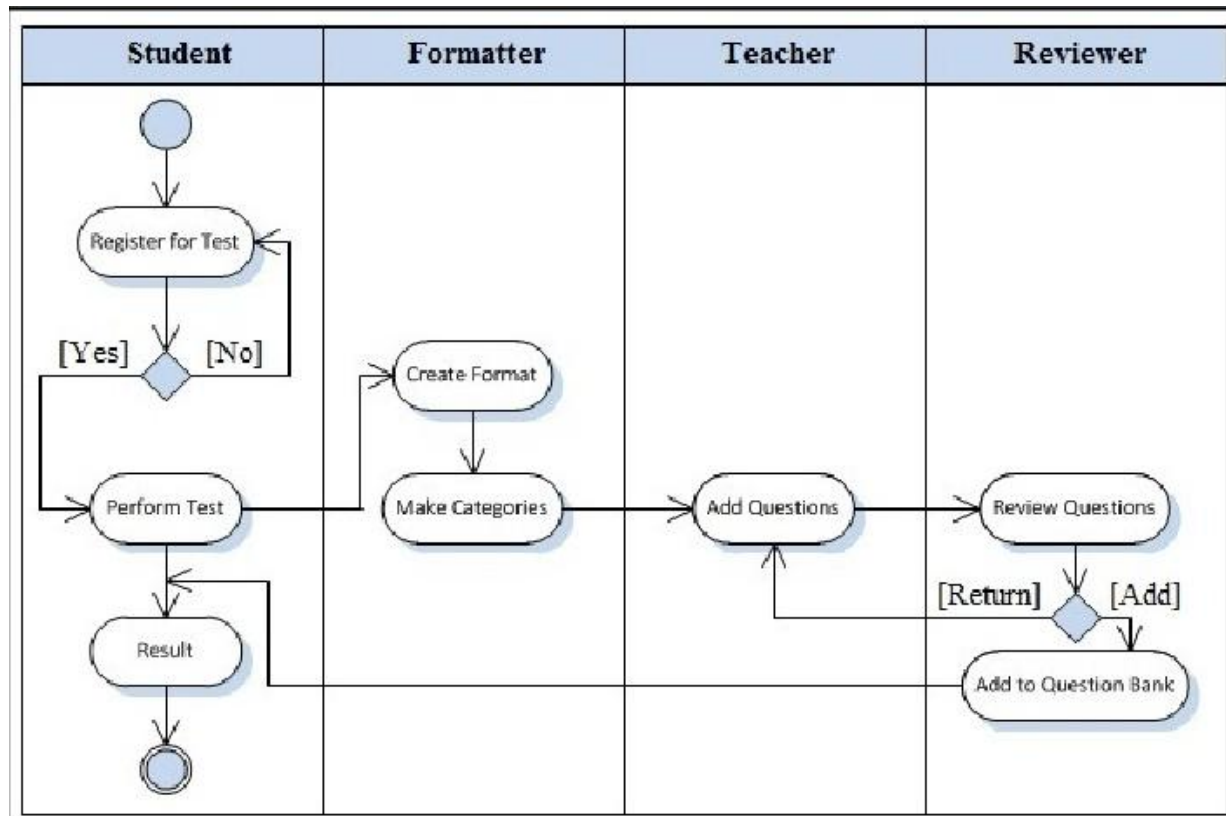
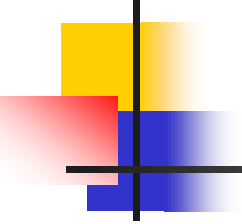


Activity diagrams



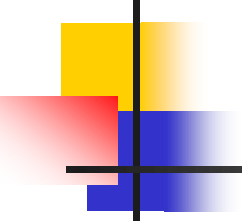


Drawing Activity Diagrams

Purpose:

- To model a business process
- To describe a function of a system represented by a use case
- To describe the logic of an operation

Activity diagrams shows **sequential** and **parallel** activities in a process



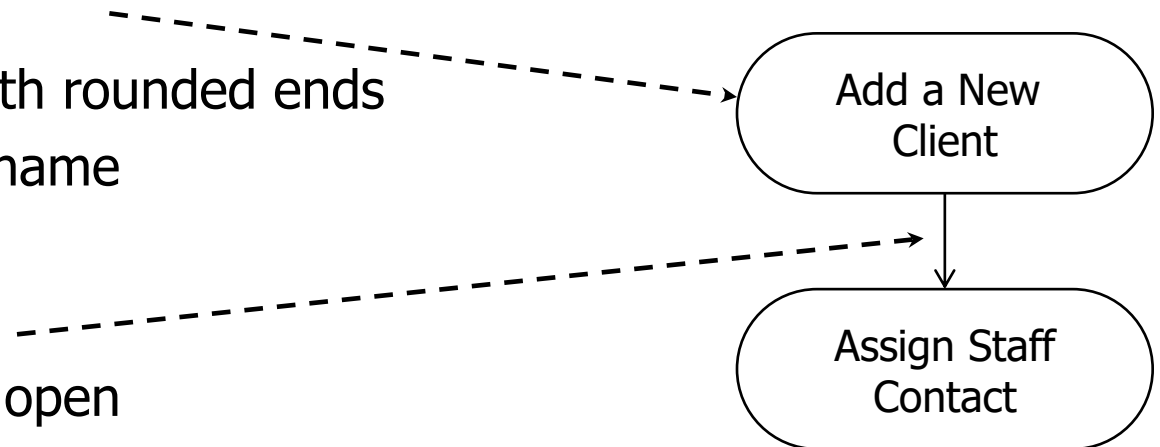
Notation of Activity Diagrams

- **Activities**

- rectangle with rounded ends
- meaningful name

- **Transitions**

- arrows with open arrowheads



Notation of Activity Diagrams

- **Start state**

- black circle

- **Decision points**

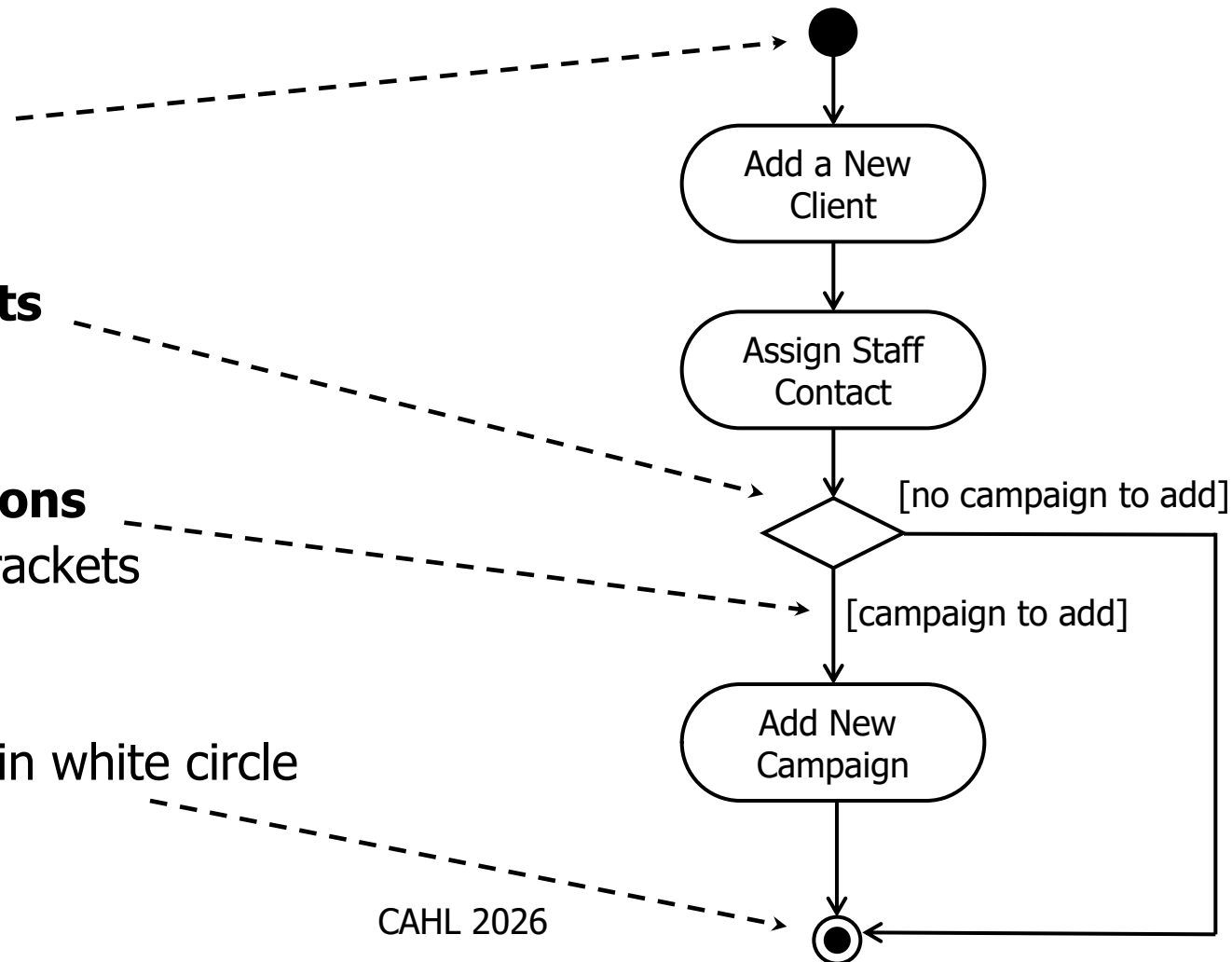
- Diamond

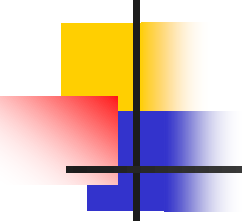
- **Guard conditions**

- in square brackets

- **Final state**

- black circle in white circle





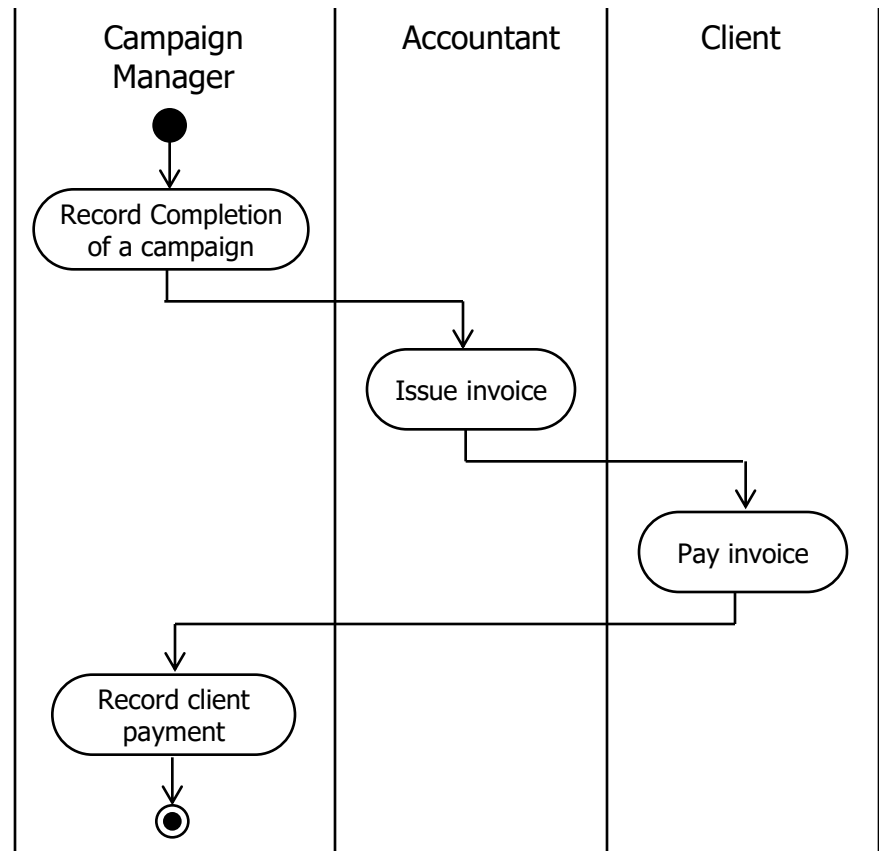
Drawing Activity Diagrams

- Are all the activities carried out by the same person, organisation or department?
- If not, then add swim lanes to show the responsibilities
- Name the swim lanes
- Show each activity in the appropriate swim lane

Notation of Activity Diagrams

■ Swim lanes

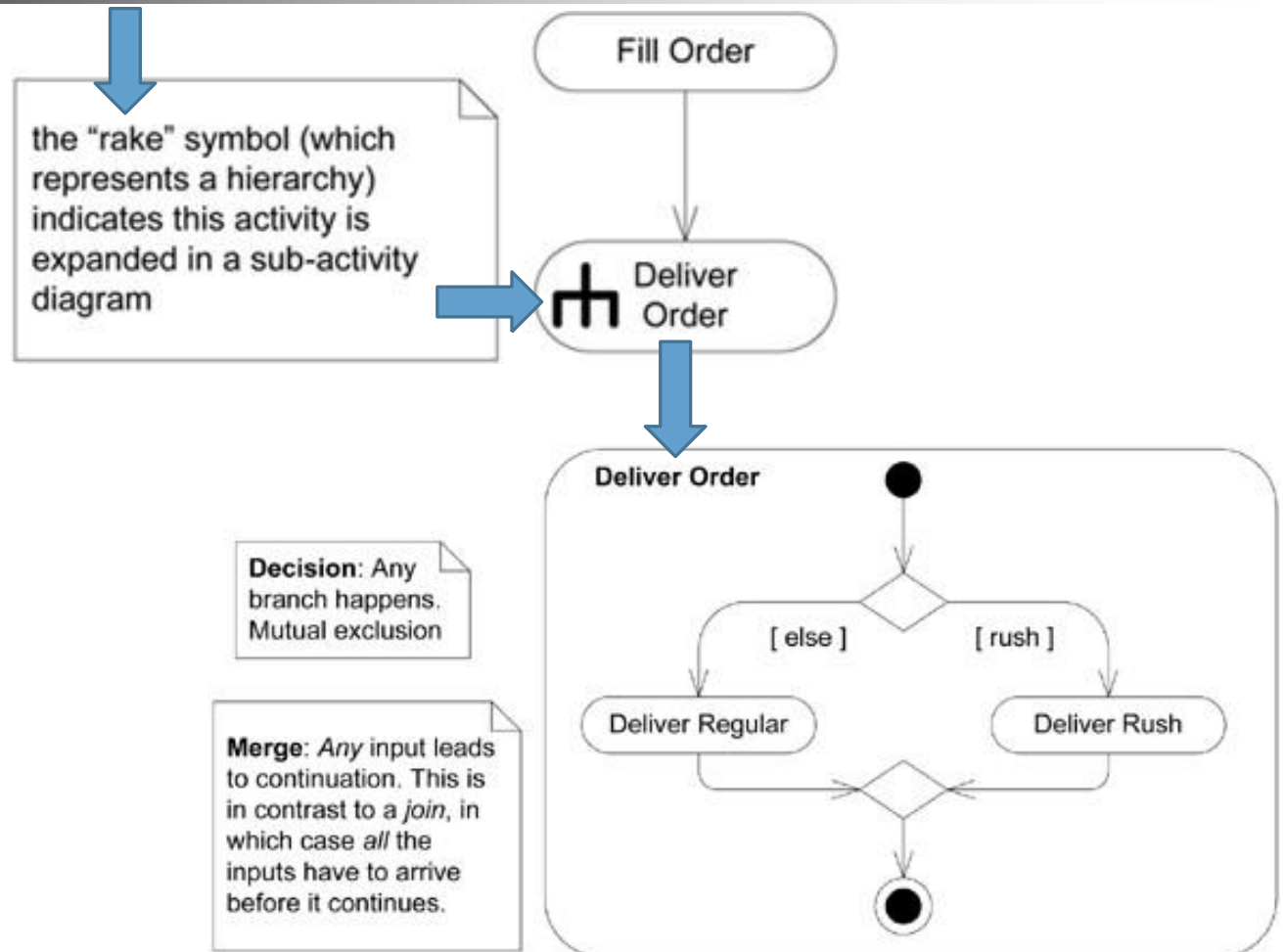
- vertical columns
- labelled with the person, organisation or department responsible for the activities in that column



Sub-activity diagrams

Symbols:

- Rake
- Decision
- Merge

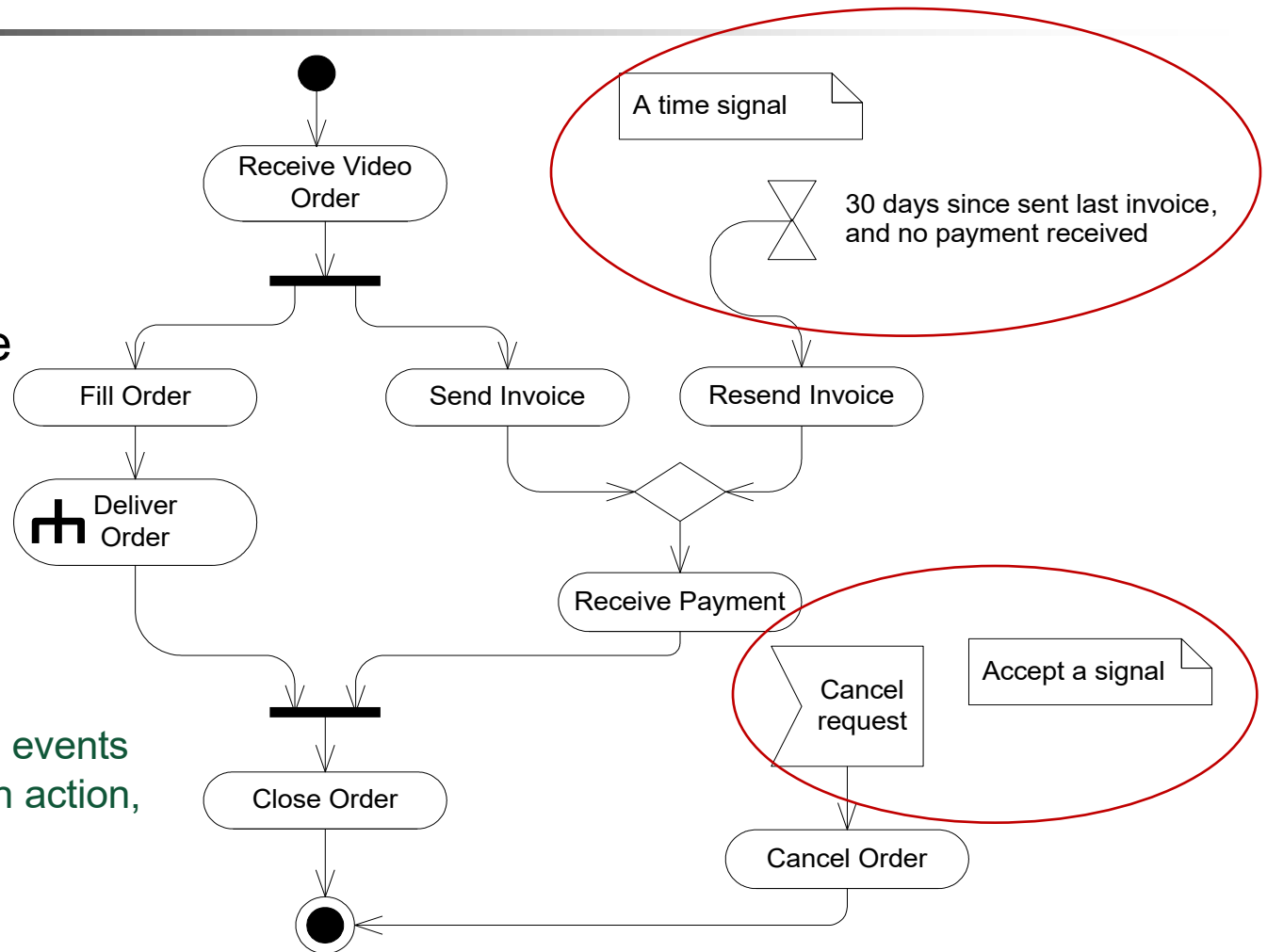


More UML Activity Diagram notation

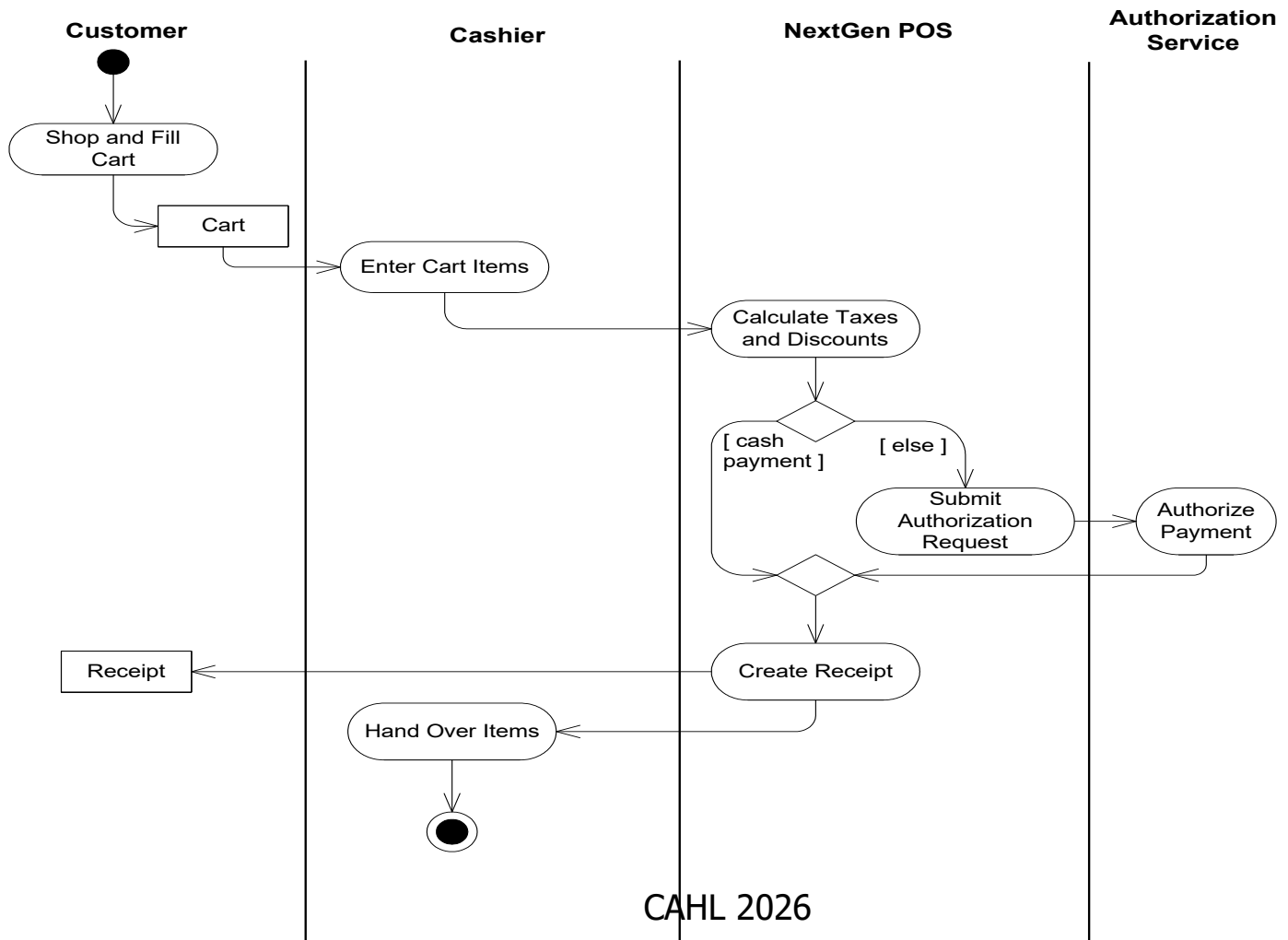
Symbols:

- Time event
- Send and Receive
 - Event signals

When you need to model events such as time triggering an action, or a cancellation request



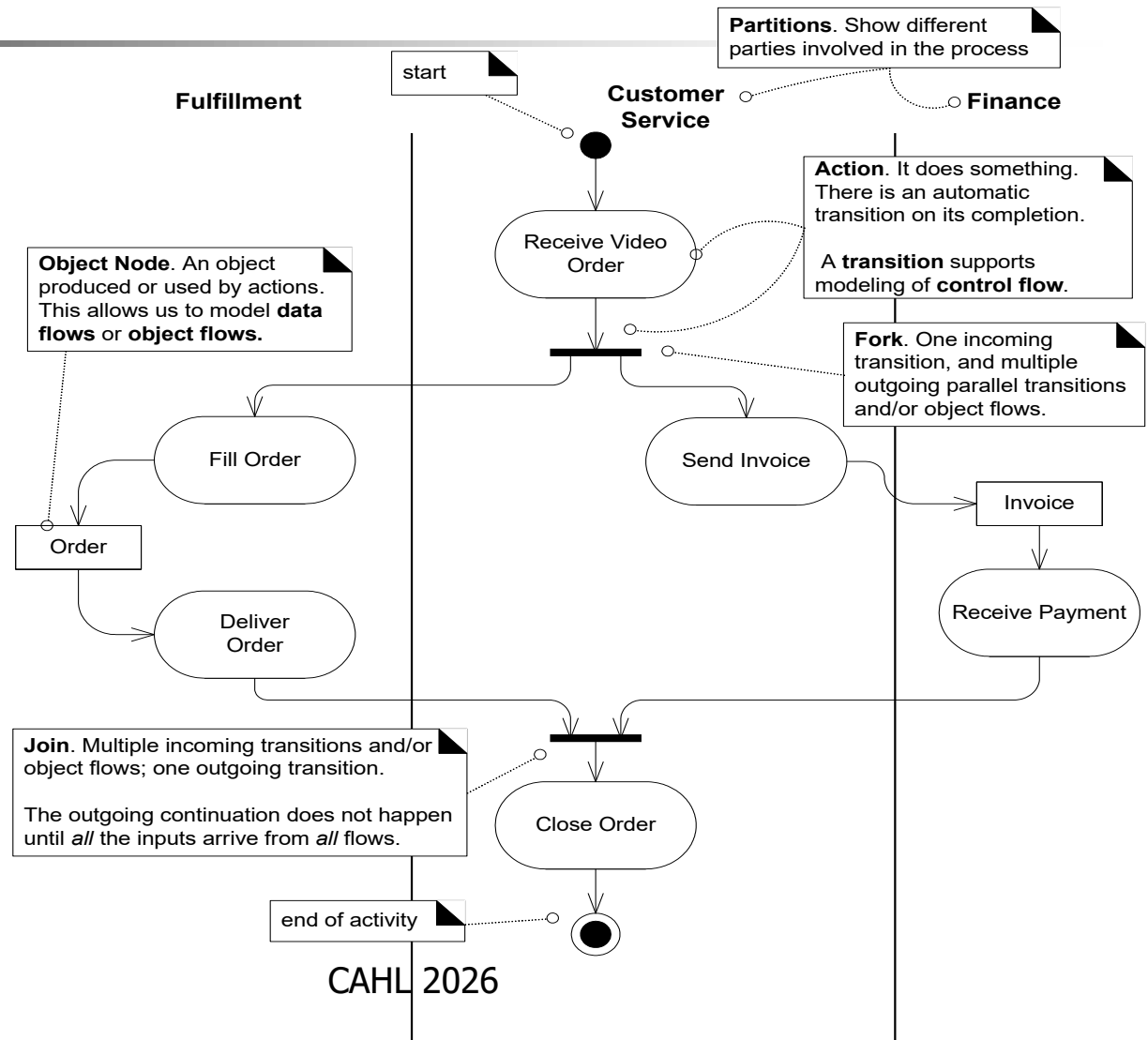
Process sale use case

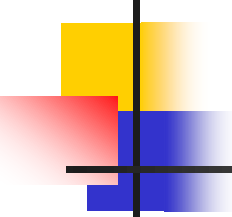


Activity diagrams with forks and object Nodes

Basic notations:

- Action
- Partition
- Fork
- Join
- Object node
- Swimlane



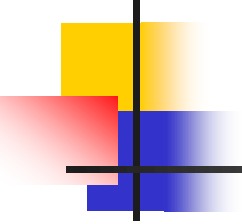


Activity Diagrams in the development process

- Activity diagrams are especially valuable in the **Business Modeling** discipline
- Activity diagrams can help to understand and communicate the structure and the dynamics of the organization

Guidelines:

- *Activity diagrams are best suited for complex processes. Use case text suffices for simple processes*
- *Take advantage of the possibility to make sub-activity diagrams*
- *Strive to make the level of abstraction of action nodes roughly equal within a diagram*



UML Notationer – Samlet oversigt

- **Startpunkt** – Fyldt sort cirkel
- **Slutpunkt** – Sort cirkel med hvid kant
- **Aktivitet** – Afrundet rektangel med handlingsnavn
- **Transition** – Pil fra én aktivitet til næste
- **Beslutningspunkt/Merge** – Rudeformet symbol (decision node)
- **Fork / Join** – Tykke streger der viser parallelle forløb eller samling
- **Sub-diagram (rake)** – Aktivitetsboks med lille 'rake'-ikon i hjørnet
- **Objekt-node** – Rektangel der viser data der oprettes eller bruges
- **Tidsbegivenhed / signal** – Bruges til at modellere hændelser eller tidsudløste handlinger
- **Swimlane** – Lodrette kolonner opdelt efter ansvar (aktør eller afdeling)